02- HEAT CAPACITY OF METALS

PRINCIPLE

Heated specimens are placed in a calorimeter filled with water at low temperature. The heat capacity of the specimen is determined from the rise in the temperature of the water.

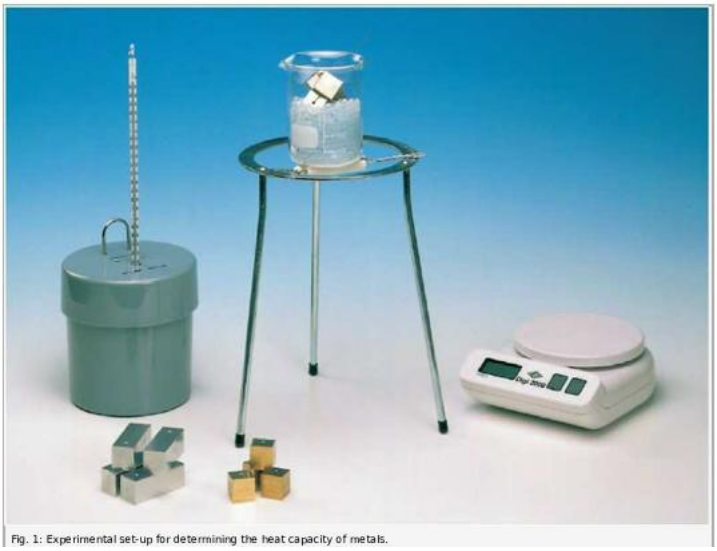

Fig. 1: Experimental set-up for determining the heat capacity of metals.

TASKS

1. To determine the heat capacity of the calorimeter by filling it with hot water and determining the rise in temperature.
2. To determine the specific heat capacity of aluminium, iron and brass.
3. To verify Dulong Petit's law with the results of these experiments.

THEORY

The heat capacity C of a substance is defined as the quotient of the quantity of heat absorbed δQ and the change in the temperature dT:

$$C = \frac{\delta Q}{dT} \tag{1}$$

and is the proportional to the mass the heated substance.

$$c = \frac{C}{m} \tag{2}$$

is the specific heat capacity.

The quantity of heat absorbed δQ depends on the conditions prevalling as the temperature rises, and a differentiation is made in particular between heat capacity C_V at constant volume V and heat capacity C_P at constant pressure p.

In accordance with the First Law of Thermodynamics (U=Internal Energy).

$$\delta Q = dU + pdV \tag{3}$$

is always greater than. In the case of solids, the change in volume is so small that we can write

$$C_P \cong C_V$$

Can abe calculated from the change in internal energy with temperature in accordance with (1) and (3):

$$C_V = \left(\frac{\partial U}{\partial T}\right)_V \tag{4}$$

The internal energy U in a solid is essentially the result of lattice vibrations caused by heat.

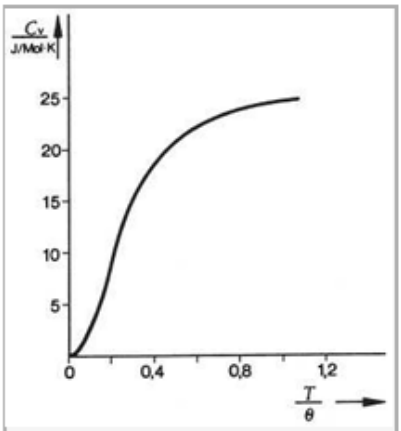

Fig. 2: Molar heat capacity of a solid in accordance with Debye's approximation.

According to Debye's theory, which considers lattice vibrations up to a limiting frequency, the heat capacity is given by

$$C_V(T) = 3N_k\left(\frac{T}{\theta}\right).3\int_0^{\theta/T}\left(\frac{z^4e^z d_z}{(e^z-1)^2}\right) = 3Nk.D\left(\frac{T}{\theta}\right) \quad (5)$$

where

$z = {}^{h}/_{kT}$,

$0 = {}^{hvD}/_{k}$, called the Debye temperature

h= Planck'S constant,

k= Boltzmann constant,

N= number of atoms in the volume considered.

$D\left(\frac{T}{\theta}\right)$ is called the Debye function.

For large values of $\left(\frac{T}{\theta}\right)$ the upper integration limit is small, the he integrand can be expanded and we obtain the law of Dulong-Petit:

$$C_V(T) = 3N_k$$

We thus obtain the molar heat capacity

$$C_m = 3N_L.k = 3R = 24.94\frac{J}{K} \quad (6)$$

where N_L is the Loschmidt number and R the gas constant.

Debye temperature:
Aluminium: 419 K

Copper: 335 K

Iron: 462 K

Zinc: 100 K

For the evaluation, the heat capacity is assumed to be constant in the temperature range considered.

The mixture (heat sharing) temperature ϑ_m is determined by extreapolating the curves plotted, as the sketch in Fig.3 shows.

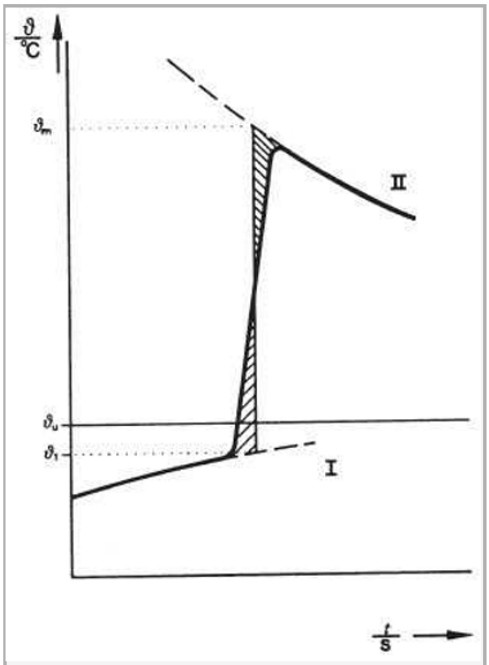

Fig. 3: Determination of the initial temperature ϑ_1 and the mixture temperature ϑ_m in the method of mixtures experimentwhen heat loss takes place. ϑ_u = temperature of the surrounding atmosphere.

The temperature before and after heat sharing are not constant because of the Exchange of heat with the surroundings; they approach the ambient temperature exponentially.

Fort he evaluation we draw a straight line parallel to the temperature axis in the graph of the temperature rise, so that the shaded parts in Fig.3 are euqal in area.

ϑ_m and ϑ_1 are read off at athe points where the straight line intersects the extended partial curves I and II.

1. First of all, the heat capacity of the calorimeter is determined:

$$C_k = c_w . m_w \frac{\vartheta_w - \vartheta_m}{\vartheta_w - \vartheta_k} \tag{7}$$

where ϑ_k is the temperature of the calorimeter before experiment, ϑ_m is the temperature of the mixture, ϑ_w is the temperature of the hot water, m_w is the mass of the water and c_w= 4.187 J/gk, the specific heat capacity of water.

The average value from several measurements (with the vessel half full) is

$$C_k = (66 \pm 1)\, J/K$$

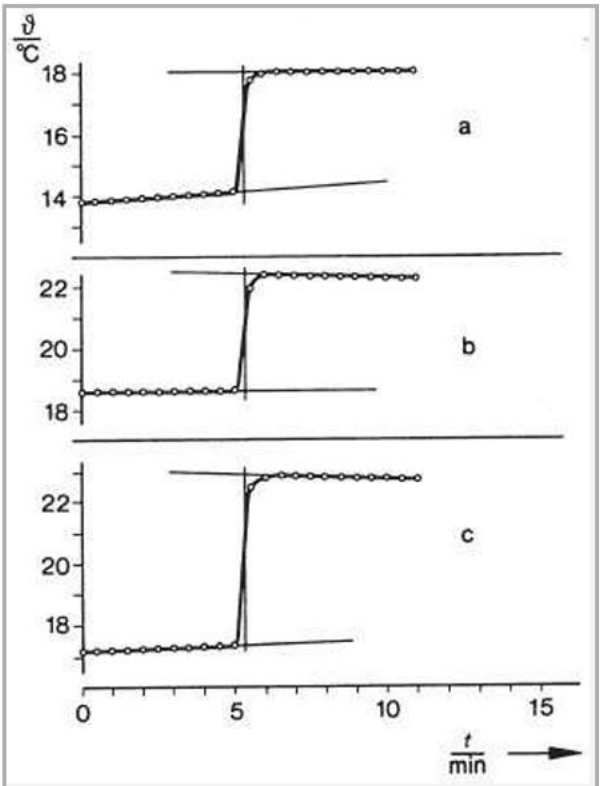

Fig. 4: Temperature as a function of time in the method of mixtures experiment a) steel, b) brass, c) aluminium.

2. The test pieces of mass $m_p$are heated in the boiling water. The boiling point ϑ_2 is determined as a function of the atmospheric pressure p:

$$\vartheta_2 = 100 + 0.0276(p - 1013hPa) - 0.000017(p - 1013hPa) \quad (8)$$

where p is in hPa and ϑ_2 in °C.

The specific heat capacity of the material from which the test pieces are made is obtained from the energy balance as:

$$c = \frac{c_w . m_w + C_k\,(\vartheta_m - \vartheta_1)}{m_p\,(\vartheta_2 - \vartheta_m)} \quad (9)$$

The values of molar heat capacity of brass and aluminium as measured in the experiment agree well with the values from Dulong and Petit's law.

The specific heat capacity of steel depends very much on the composition of the steel.

	$\frac{\vartheta_2}{°C}$	$\frac{m_p}{g}$	$\frac{m_w}{g}$	$\frac{c}{J/g{\cdot}K}$	$\frac{C_m}{J/Mol{\cdot}K}$
Brass	99.8	120.8	214.6	0.387	24.9
Aluminium	99.3	90.6	251.1	0.897	24.2
Steel	99.3	117.2	257.4	0.469	

SUMMARY

Specific Heat

Specific heat (also known as heat capacity) is the amount of heat required to change the temperature of 1 kilogram (1000 grams) of a substance by 1 °C, whether by adding heat or removing heat. To determine a substance's specific heat, heat must be added or removed. One effective method for this measurement is to bring the substance into contact with another substance at a different temperature, allowing them to reach thermal equilibrium. This approach is referred to as the "***mixing method."***

There are two primary rules, known as ***calorimetric laws***, that are applied when solving problems related to heat calculation and temperature change:

a) When substances with different temperatures are in contact, the hotter substances will cool down, while the colder substances will warm up until thermal equilibrium is achieved. In summary:

Heat Absorbed = Heat Given

b) The amount of heat required for an event to occur is equal to the amount of heat released when the opposite of that event occurs. The formula for the amount of heat needed to raise the temperature of an object with mass m from temperature t_1 to $t_2$is expressed as:

$$Q = m \cdot c(t_2 - t_1) = m \cdot c \cdot \Delta t$$

Atomic Heat

Atomic heat refers to the amount of heat needed to raise the temperature of one gram of an atom of a substance by 1 °C, which is denoted by C.

$$C = A \cdot c$$

where A is the atom weight.

Dulong-Petit Law

According to Dulong-Petit law, the atomic heat capacities of metals are approximately 6.4. This law allows for the determination of the heat capacities of all metals with known atomic weights.

Before determining the heat capacities of various materials, it is essential to first

establish the heat capacity of the calorimeter used in the experiment.

Table I:

ϑ_k(°C)	ϑ_w(°C)	m_w	ϑ_m (°C)	C_k(J/°K)

Equipment

Calorimeter container, metal parts, thermometer, heating tube, beaker, scale, fishing line, K-type thermocouple, multimeter.

Set-up and Prodecure:

Part I

1. Measure the calorimeter temperature before the experiment.
2. Pour (m_w) 300ml of water with hot water (approx. ϑ_w≈50°C) into the calorimeter.
3. Repeat the experiment to increase the accuracy of the measurement. Keep ready a vessel containing water at room temperature, so that the calorimeter can quickly be returned to room temperature after the measurement. Leave this water in the calorimeter for approx. 5 minutes and then empty and dry it.
4. Start the next measurement after another five minutes.
5. Record the values you obtained in Table I.

$$C_{k_{avg}} = \qquad j/K$$

Part II

1. Weigh each metals pieces to obtain their mass m_p.
2. Put a layer of glass beads about 3 cm deep in the bottom of the beaker so that the metal specimens do not touch it when they are being heated.
3. Fill the calorimeter with cold water of a known mass.
4. Leave the specimens in the boiling water for about 10 minutes, then drain them quickly and put them into the calorimeter.

5. Find the specific heat capacities for brass, steel, and aluminum and record the results in Table II.

Table II:

	ϑ_k(°C)	ϑ_2(°C)	ϑ_m(°C)	$m_p(gr)$	$m_w(gr)$	$c(J/gr.°K)$
Brass						
Aluminium						
Iron						

EXPERIMENT-RELATED QUESTIONS

1. What are heat and temperature? How can we distinguish these two physical quantities from each other?

2. Debye's theory can accurately explain heat capacity at low temperatures. Is Debye's theory valid at high temperatures? Research this.

3. Another theory proposed to explain heat capacity in solids is known as Einstein's theory. Re-establish heat capacity using Einstein's theory. Draw the heat capacity expression you obtained as a function of temperature and compare it with the results of Debye's theory.

4. What is latent heat? Is there a relationship between latent heat and heat capacity? Research this.

5. In which of the reversible and irreversible processes is the heat change greater?

6. What are the laws of thermodynamics? Explain them.

7. What physical consequences would you expect if a material had no heat capacity?

8. How can we increase the heat capacity of materials, mixtures, liquids, etc.? Have you ever wondered?

9. What is a solar pool? Research it.

10. Can a material with increased heat capacity be used in industry, manufacturing, and for the benefit of society?

11. Interpret the graph you drew. Are the results as you expected?

REFERENCES

1. PHYWE series of publications, Laboratory Experiments